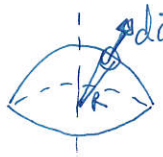
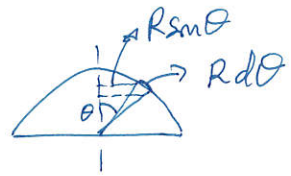
P1.61

$\vec{a} = \int d\vec{a}$

a)



$d\vec{a} = \hat{r}\, R^2 \sin\theta\, d\theta\, d\varphi$



$\hat{r} = \sin\theta\cos\varphi\,\hat{x} + \sin\theta\sin\varphi\,\hat{y} + \cos\theta\,\hat{z}$

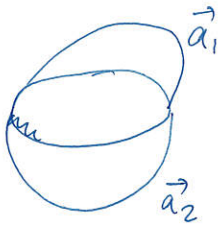
$$\Rightarrow \vec{a} = R^2\Big(\hat{x}\underbrace{\int \sin\theta\cos\varphi\cdot\sin\theta\, d\theta\, d\varphi}_{=0} + \hat{y}\underbrace{\int \sin\theta\sin\varphi\cdot\sin\theta\, d\theta\, d\varphi}_{=0}$$

$$+ \hat{z}\int \cos\theta\cdot\sin\theta\, d\theta\, d\varphi\Big) = \hat{z}\, 2\pi R^2 \int_0^{\pi/2} \cos\theta\sin\theta\, d\theta =$$

$$= \hat{z}\, 2\pi R^2\, \frac{1}{2}\sin^2\theta\Big|_0^{\pi/2} = \pi R^2\,\hat{z}.$$

b) $\oint_{\text{closed surface}} d\vec{a} = \int \vec{\nabla}(1)\, d^3\vec{r} = 0$ using (P1.60a)

c)



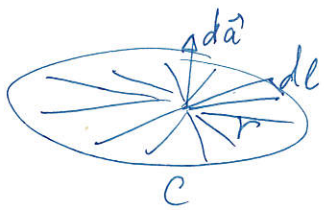
suppose $\vec{a}_1 \neq \vec{a}_2$

form closed surface by constructing

$\vec{a}_1 - \vec{a}_2 = \oint_{\text{closed}} d\vec{a} \neq 0$. which contradicts part (b)

$\Rightarrow \vec{a}_1 = \vec{a}_2$.

d)



$d\vec{a} = \frac{1}{2}\,\vec{r}\times d\vec{\ell}$

$\Rightarrow \vec{a} = \frac{1}{2}\oint_C \vec{r}\times d\vec{\ell}$

e) P1.60e says: $\int_S \vec{\nabla} T \times d\vec{a} = -\oint_C T\, d\vec{\ell}$

let $T = \vec{c}\cdot\vec{r}$ where $\vec{c} = \text{const} \Rightarrow \vec{\nabla} T = \vec{\nabla}(\vec{c}\cdot\vec{r}) =$

$= \left(\hat{x}\frac{\partial}{\partial x} + \hat{y}\frac{\partial}{\partial y} + \hat{z}\frac{\partial}{\partial z}\right)(c_x x + c_y y + c_z z) = \hat{x}c_x + \hat{y}c_y + \hat{z}c_z = \vec{c}$

$\Rightarrow \int_S \vec{c}\times d\vec{a} = -\oint_C \vec{c}\cdot\vec{r}\, d\vec{\ell} \Rightarrow \oint_C \vec{c}\cdot\vec{r}\, d\vec{\ell} = -\vec{c}\times\int d\vec{a} = -\vec{c}\times\vec{a}$

$= \vec{a}\times\vec{c}$.

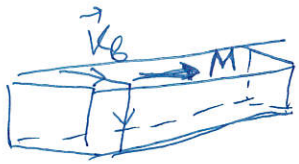
Then (5.81) $\vec{A}(\vec{r}) = \frac{\mu_0 I}{4\pi r^2} \oint (\hat{r} \cdot \vec{r}\,') d\vec{l}\,'$

but $\oint \hat{r} \cdot \vec{r}\,' d\vec{l}\,' = -\hat{r} \times \int d\vec{a}\,' = -\hat{r} \times \vec{a}\,' = \vec{a}\,' \times \hat{r}$

$$\Rightarrow \vec{A}(\vec{r}) = \frac{\mu_0 I}{4\pi r^2} \vec{a}\,' \times \hat{r} = \frac{\mu_0}{4\pi} \frac{I \vec{a}\,' \times \hat{r}}{r^2} = \frac{\mu_0}{4\pi} \frac{\vec{m} \times \hat{r}}{r^2}$$

where $\vec{m} = I \vec{a}\,' = I \int d\vec{a}\,'$ = mag. dipole moment.

P6.10

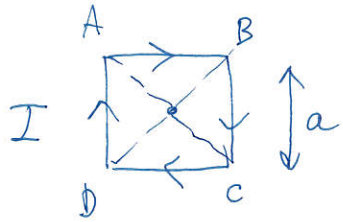


Iron rod: magnetization $\vec{M} = \text{const}$

$\Rightarrow$ surface current $\vec{K}_b = \vec{M} \times \vec{n}$ circles around the rod surface.

$\Rightarrow$ just as a solenoid $B_{rod} = \mu_0 K_b = \mu_0 M$.

Magnetic field of a square loop:



$$\vec{B}_{sq} = \vec{B}_{AB} + \vec{B}_{BC} + \vec{B}_{CD} + \vec{B}_{DA}$$

from eq.(5.35) $B_{AB} = \dfrac{\mu_0 I}{4\pi\left(\frac{a}{2}\right)}\left(\sin\frac{\pi}{4} - \sin\left(-\frac{\pi}{4}\right)\right) = \dfrac{\mu_0 \sqrt{2}\, I}{2\pi a}$

(into page)

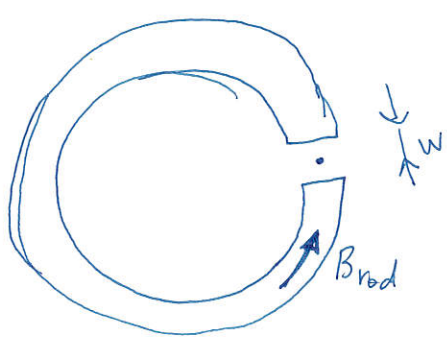
$$B_{sq} = 4 B_{AB} = \frac{\mu_0 I}{\pi a} 2\sqrt{2}$$

But $I = K_b \cdot W$ (W = gap in the original problem)

$= M \cdot W$

$$B_{gap} = B_{rod} - B_{sq} = \mu_0 M - \frac{\mu_0 M W}{\pi a} 2\sqrt{2} =$$

$$= \mu_0 M \left(1 - \frac{2\sqrt{2}\, W}{\pi a}\right).$$



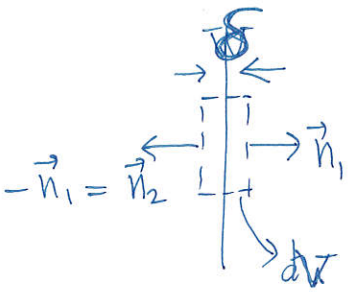
P6.15 $\vec{H} = -\vec{\nabla} W$ if $\vec{J}_f = 0$

$$\Rightarrow \nabla^2 W = \underbrace{\vec{\nabla}\cdot\vec{M}}_{\text{effective charge density}} \quad (1)$$

<u>Boundary conditions</u> $\underbrace{W(\vec{r}_2)}_{\text{outside}} - \underbrace{W(\vec{r}_1)}_{\text{inside}} = \int_{r_1}^{r_2} \vec{\nabla} W \cdot d\vec{r} = -\int_{r_1}^{r_2} \vec{H}\cdot d\vec{r} = 0$

for $\vec{r}_1 \to \vec{r}_2$.

Integrate (1) over boundary region



take the limit $\delta \to 0$:

$$\int \nabla^2 W \, dV = \oint \vec{\nabla} W \cdot d\vec{a} = \Big(\underbrace{\vec{\nabla} W \cdot \vec{n}_1}_{\text{just outside}} - \underbrace{\vec{\nabla} W \cdot \vec{n}_1}_{\text{just inside}}\Big) \text{ Area} =$$

$$= \int \vec{\nabla}\cdot\vec{M} \, dV = \oint \vec{M}\cdot d\vec{a} =$$

$$= \Big(\underbrace{\vec{M}\cdot\vec{n}_1}_{\text{outside}} - \underbrace{\vec{M}\cdot\vec{n}_1}_{\text{inside}}\Big) \text{ Area}$$

$$\Rightarrow \vec{\nabla} W \cdot \hat{n}\Big|_{\text{outside}} - \vec{\nabla} W \cdot \hat{n}\Big|_{\text{inside}} = -\vec{M}\cdot\hat{n}\Big|_{\text{inside}}$$

(because $\vec{M}_{\text{outside}} = 0$)

Thus we have:

BC
$$\begin{cases} 1)\ W_{out} = W_{in} \\ 2)\ -\dfrac{\partial}{\partial n} W_{out} + \dfrac{\partial}{\partial n} W_{in} = \vec{M}\cdot\hat{n} \end{cases}$$

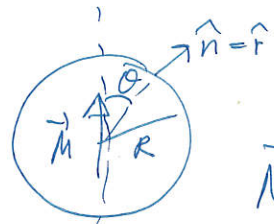
where $\dfrac{\partial}{\partial n} W = \vec{\nabla} W \cdot \hat{n}$ = normal derivative.

Away from the boundary $\nabla^2 W = 0$

$\Rightarrow$ using eq. (3.65) $\begin{cases} W_{in}(r,\theta) = \sum_\ell A_\ell r^\ell P_\ell(\cos\theta) \\ W_{out}(r,\theta) = \sum_\ell \frac{B_\ell}{r^{\ell+1}} P_\ell(\cos\theta) \end{cases}$

Apply BC: $(r = R)$

1) $\sum_\ell A_\ell R^\ell P_\ell(\cos\theta) = \sum_\ell \frac{B_\ell}{R^{\ell+1}} P_\ell(\cos\theta)$



$\vec{M} \cdot \hat{n} = \vec{M} \cdot \hat{r} = M\cos\theta$

2) $\sum_\ell \frac{(\ell+1) B_\ell}{R^{\ell+2}} P_\ell(\cos\theta) + \sum_\ell \ell A_\ell R^{\ell-1} P_\ell(\cos\theta) = M\cos\theta$

$\Rightarrow A_\ell = \frac{B_\ell}{R^{2\ell+1}}$

$\Rightarrow \sum_\ell \left(\frac{(\ell+1) B_\ell}{R^{\ell+2}} + \frac{\ell B_\ell R^{\ell-1}}{R^{2\ell+1}} \right) P_\ell(\cos\theta) = M\cos\theta$

$\sum_\ell \frac{(2\ell+1)}{R^{\ell+2}} B_\ell P_\ell(\cos\theta) = M\cos\theta \Rightarrow$ using (Table 3.1)

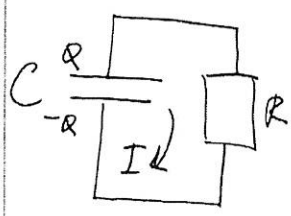
$\frac{3B_1}{R^3} = M$, $B_\ell = 0$ for all $\ell \neq 1$

$\Rightarrow A_1 = \frac{B_1}{R^3} = \frac{MR^3}{3R^3} = \frac{1}{3}M.$

$\Rightarrow W_{in}(r,\theta) = A_1 r P_1(\cos\theta) = \frac{1}{3} M r\cos\theta = \frac{1}{3} M z$

$\vec{H}_{in} = -\vec{\nabla} W_{in} = -\frac{1}{3} M \hat{z} \Rightarrow \vec{B}_{in} = \mu_0(\vec{H}_{in} + \vec{M}) = \frac{2M\hat{z}}{3}\mu_0.$

7.2



$$Q_0 = C V_0 .$$

$$V(t) = I(t) R$$

(a) $\frac{dQ}{dt} = -I$ because charge on capacitor is decreasing.

$$\Rightarrow \frac{1}{C} Q = -\frac{dQ}{dt} R \Rightarrow \frac{dQ}{dt} = -\frac{1}{RC} Q$$

$$\Rightarrow Q(t) = Q_0 e^{-t/RC} = C V_0 e^{-t/RC}$$

$$I(t) = -\frac{dQ}{dt} = \frac{C V_0}{RC} e^{-t/RC} = \frac{V_0}{R} e^{-t/RC}$$

(b) $U_0 = \frac{Q_0^2}{2C} = \frac{1}{2} C V_0^2 .$

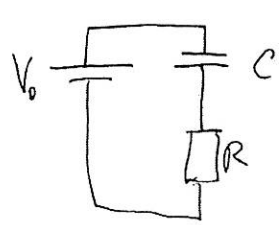
$$P = I^2 R \Rightarrow W = \int_0^\infty I^2 R \, dt = \frac{V_0^2}{R} \int_0^\infty e^{-\frac{2t}{RC}} dt =$$

$$= \frac{V_0^2}{R} \left(-\frac{RC}{2}\right) e^{-2t/RC} \Big|_0^\infty = \frac{V_0^2}{R} \left(-\frac{RC}{2}\right)(-1) =$$

$$= \frac{1}{2} C V_0^2 .$$

$$\Rightarrow W = U_0 .$$

(c)



$$V_0 = \frac{Q}{C} + IR .$$

At $t=0$ charge $Q(t=0)=0 \Rightarrow I = \frac{dQ}{dt}$

$$\Rightarrow V_0 = \frac{Q}{C} + R \frac{dQ}{dt}$$

$$\frac{dQ}{dt} = \frac{CV_0 - Q}{RC} \Rightarrow \frac{dQ}{Q - CV_0} = -\frac{1}{RC} dt \Rightarrow \ln(Q - CV_0) = \frac{-t}{RC} + \alpha$$

constant

$$\Rightarrow Q(t) = CV_0 + e^{\alpha} e^{-t/RC} \quad \text{but } Q(t=0) = 0 \Rightarrow CV_0 + e^{\alpha} = 0$$

$$\Rightarrow Q(t) = CV_0 \left(1 - e^{-t/RC}\right) ; \; I(t) = \frac{dQ}{dt} = \frac{V_0}{R} e^{-t/RC} .$$

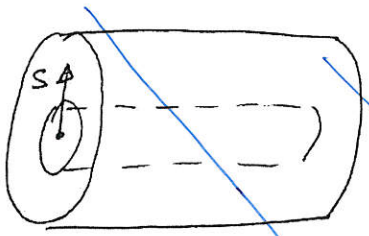
(d) Energy from the battery $= \int_0^\infty V_0 I \, dt =$

$$= \frac{V_0^2}{R} \int_0^\infty e^{-t/RC} \, dt = \frac{V_0^2}{R}(-RC) \underbrace{e^{-t/RC}\Big|_0^\infty}_{-1} = CV_0^2.$$

Since energy on the capacitor $= \frac{Q^2(t=\infty)}{2C} = \frac{(CV_0)^2}{2C} = \frac{1}{2}CV_0^2$

we observe that out of CV_0^2 delivered by the battery $\frac{1}{2}CV_0^2$ went to the capacitor, and another $\frac{1}{2}CV_0^2$ was lost on the resistor.

$\boxed{4.4}$



$$\sigma(s) = \frac{k}{s}$$

Total current through cross-section at distance s from the center

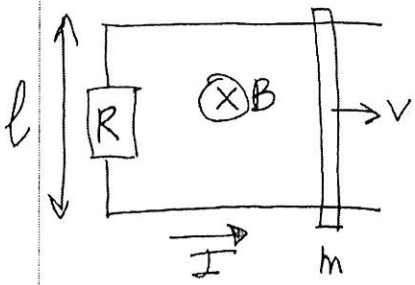
$$\left.\begin{aligned} I &= J \cdot 2\pi s \cdot L \\ J &= \sigma E = \frac{kE}{s} \end{aligned}\right\} \quad I = 2\pi k L E$$

$\Rightarrow E = \frac{I}{2\pi k L}$ in terms of the current.

$$V = -\int_b^a \vec{E} \cdot d\vec{s} = \frac{I}{2\pi k L}(b-a) = IR$$

where $R = \frac{b-a}{2\pi k L}$

7.7



$$\Phi = B\ell(x_0 + vt)$$

(a) $\mathcal{E} = -\frac{d\Phi}{dt} = -B\ell v$

$I = \frac{B\ell v}{R}$, flows counterclockwise so as to reduce flux Φ which is increasing with time.

(b) $F = I\ell B = \frac{B^2\ell^2 v}{R}$, points to the <u>left</u>. (along $\vec{I} \times \vec{B}$)

(c) $ma = -\frac{B^2\ell^2 v}{R}$, $a = \frac{dv}{dt}$

$$\frac{dv}{dt} = -\frac{B^2\ell^2}{mR} v \Rightarrow v(t) = v_0 e^{-\frac{B^2\ell^2}{mR}t}$$

(d) $W = \int_0^\infty I^2 R\,dt = R\left(\frac{B\ell}{R}\right)^2 v_0^2 \int_0^\infty e^{-2ct}\,dt = \frac{B^2\ell^2}{R} v_0^2 \left(\frac{-1}{2c}\right)(-1) =$

$$c = \frac{B^2\ell^2}{mR}$$

$$= \frac{B^2\ell^2}{R} \frac{v_0^2}{2\frac{B^2\ell^2}{mR}} = \frac{1}{2} m v_0^2 \;!$$